Mathematics for ML: Functions & Limits

Yogesh Haribhau Kulkarni

Outline

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About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- Feel free to follow me at:
  - Github (github.com/yogeshhk)
  - LinkedIn (www.linkedin.com/in/yogeshkulkarni/)
  - Medium (yogeshharibhaukulkarni.medium.com)
  - Send email to yogeshkulkarni at yahoo dot com

Office Hours: Saturdays, 2 to 5pm (IST); Free-Open to all; email for appointment.

About Content

- Bare essential mathematics needed for Machine/Deep Learning
- Just to get you started.
- Each field within Machine/Deep Learning can go extremely deep.
- This is to give you enough foundation needed.

About Me

- I am not a mathematician.
- And this course won't turn YOU into one!!! (if you are not already)
- But, I will surely attempt to make you get interested in the learning, more.

Calculus

- Algebra deals with finite processes.
- Calculus deals with infinitesimal processes.
- Limiting situations.
- Computers can not handle continuity, need to approximate.

Main parts of Calculus

- Functions
- Limits
- Derivatives
- Integration
- Gradients
- ...

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Basic Idea

- Approximate/Divide.
- Compose the total.
- Reduce approximation to infinitesimally small and compute.
- Get the actual results with infinitesimally small tending to 0.

Basic Idea: Example

- We know, Circumference of circle is $2\pi r$
- What's the equation of Area?

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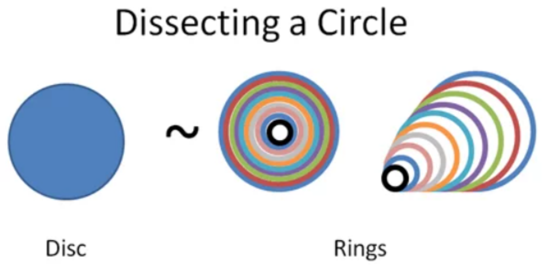
Basic Idea: Example

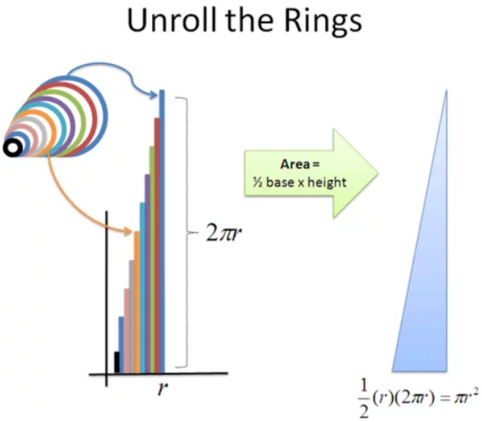
Divide the circle into concentric strips, of *small* widths.



(Ref: Lesson 1: 1 Minute Calculus: X-Ray and Time-Lapse Vision - Better Explained)

Basic Idea: Example

Compose to get the desired Area



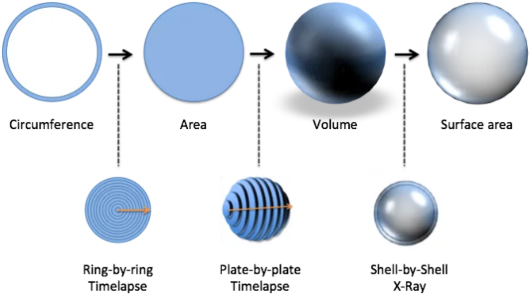
(Ref: Lesson 1: 1 Minute Calculus: X-Ray and Time-Lapse Vision - Better Explained)

Basic Idea: Example

- Still approximate, right?
- Reduce the approximation ie width to 'infinitesimally' small to get accurate results.
- This is called as . . . ??

(Ref: Lesson 1: 1 Minute Calculus: X-Ray and Time-Lapse Vision - Better Explained)

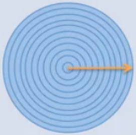
Basic Idea: Example



- Circumference with widths giving Area
- Area slices with widths giving Volume
- Slice places widths composed to give Surface Area

(Ref: Lesson 3: Expanding Our Intuition - Better Explained)

Basic Idea: Mathematical Notations

Strategy	Visualization	Step-by-Step Layout	Single Step Zoom
Ring-by-ring Timelapse			

(Ref: Lesson 4: Learning The Official Terms - Better Explained)

Basic Idea: Mathematical Notations

Intuitive Concept	Formal Name	Symbol
X-Ray (split apart)	Take the derivative (derive)	$\frac{d}{dr}$
Time-lapse (glue together)	Take the integral (integrate)	$\int$
Arrow direction	Integrate or derive "with respect to" a variable.	*dr* implies moving along *r*
Arrow start/stop	Bounds or range of integration	$\int_{start}^{end}$
Slice	Integrand (shape being glued together, such as a ring)	Equation, such as $2\pi r$

(Ref: Lesson 4: Learning The Official Terms - Better Explained)

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Basic Idea: Algebra vs Calculus

Operation	Example	Notes
Division	$\frac{y}{x}$	Split whole into identical parts
Differentiation	$\frac{d}{dx}y$	Split whole into (possibly different) parts
Multiplication	$y \cdot x$	Accumulate identical steps
Integration	$\int y\, dx$	Accumulate (possibly different) steps

(Ref: Lesson 6: Improving Arithmetic And Algebra - Better Explained)

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Basic Idea: More Formulas

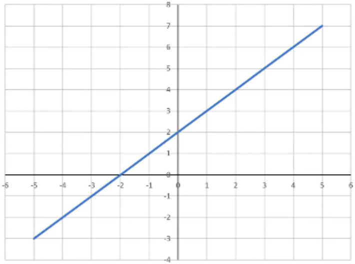
Algebra	Calculus
$distance = speed \cdot time$	$distance = \int speed\, dt$
$speed = \dfrac{distance}{time}$	$speed = \dfrac{d}{dt} distance$
$area = height \cdot width$	$area = \int height\, dw$
$weight = density \cdot length \cdot width \cdot height$	$weight = \iiint density\, dx\, dy\, dz$

(Ref: Lesson 6: Improving Arithmetic And Algebra - Better Explained)

Functions

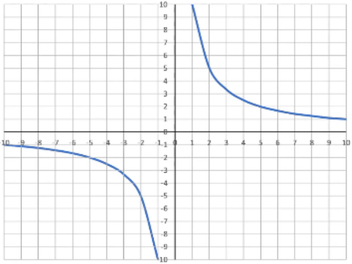
Functions

- Takes many inputs and produces one output
- $f(x) = x + 2$
- $f(3) = ?$
- This function is a line, with y is same as $f(x)$



Functions

- Another example : $g(x) = 10/x$
- $g(5) = ?$
- $g(0) = \textit{undefined}$
- Set of numbers for which the function is defined is called “Domain”.
- For g domain is $\{x \in \mathbb{R} | x \neq 0\}$



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Exercise

$h(x) = 2\sqrt{x}, x \geq 0$
Note: Although 'domain' is infinite, for plotting, need to restrict it to some range.

```python
import numpy as np
from matplotlib import pyplot as plt

def h(x):
    if x >= 0:
        return 2 * np.sqrt(x)

x = range(-100, 101)
y = [h(a) for a in x]

plt.xlabel('x')
plt.ylabel('h(x)')
plt.grid()

plt.plot(x,y, color='purple')
plt.plot(0,h(0.01), color='purple', marker='o')
plt.show()
```

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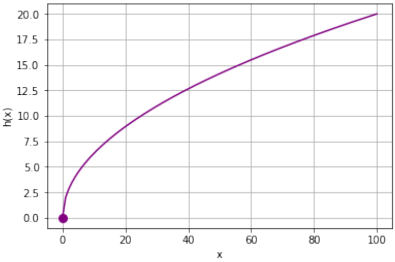
Plot



Limits

Limits, the Foundations Of Calculus

What's this? ...

Let x approach 0, but not get there, yet we'll act like it's there ...

What's going on ...

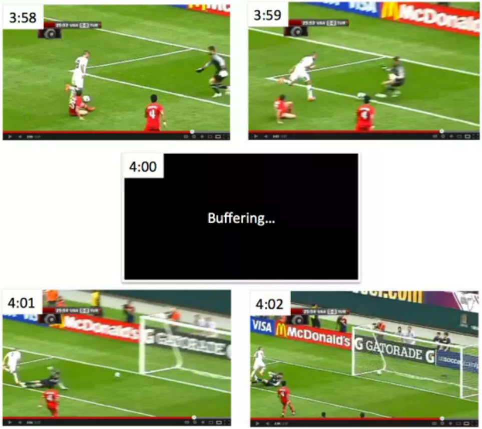
(Ref: An Intuitive Introduction To Limits - Better Explained)

Learnings

- **What is a limit?**: Our best **prediction** of a point we didn't observe.
- **How do we make a prediction?**: Zoom into the neighboring points. If our prediction is always in-between neighboring points, no matter how much we zoom, that's our estimate.
- **Why do we need limits?**: Math has "black hole" scenarios (dividing by zero, going to infinity), and limits give us an estimate when we can't compute a result directly.
- **How do we know we're right?**: We don't. Our prediction, the limit, isn't required to match reality. But for most natural phenomena, it seems to.
- **Limits let us ask "What if?"**: If we can directly observe a function at a value (like x=0, or x growing infinitely), we don't need a prediction. The limit wonders, "If you can see everything except a single value, what do you think is there?".
- **Prediction**: When our prediction is consistent and improves the closer we look, we feel confident in it. And if the function behaves smoothly, like most real-world functions do, the limit is where the missing point must be.

(Ref: An Intuitive Introduction To Limits - Better Explained)

Say, you missed a frame



Can you guess/predict ball position at the missing frame?

(Ref: An Intuitive Introduction To Limits - Better Explained)

Are you sure?

- Theoretically, the ball could have been anywhere, right?
- We just don't know, but we can predict.
- The predictions agree at increasing zoom levels.
- The before-and-after agree.
- On the contrary: Imagine at 3:59 the ball was at 10 meters, rolling right, and at 4:01 it was at 50 meters, rolling left. What happened? We had a sudden jump (a camera change?) and now we can't pin down the ball's position. Which one had the ball at 4:00? This ambiguity shatters our ability to make a confident prediction.

(Ref: An Intuitive Introduction To Limits - Better Explained)

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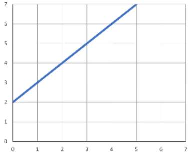
Limits

- Functions can be plotted, with input(s) on say x and output on y (higher dimension functions, not considered as of now)
- Say, $f(x) = x + 2$ is function showing distance traveled (y) at each x seconds.
- Slope gives rate of change, ie Speed.and its constant.
- It can be calculated by taking any two points on the line

$$f(x) = x + 2$$



$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

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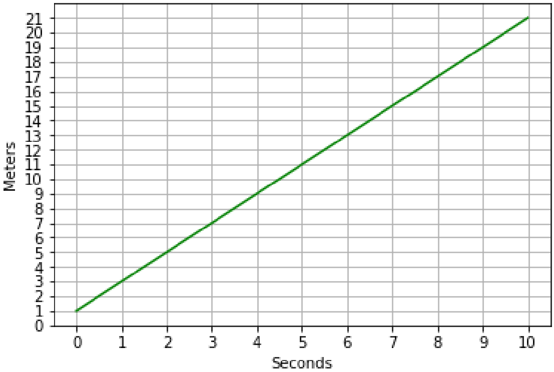
Exercise

$$q(x) = 2x + 1$$

```python
import numpy as np
from matplotlib import pyplot as plt

def q(x):
    return 2*x + 1

x = np.array(range(0, 11))

plt.xlabel('Seconds')
plt.ylabel('Meters')
plt.xticks(range(0,11, 1))
plt.yticks(range(0, 22, 1))
plt.grid()

plt.plot(x,q(x), color='green')
plt.show()
```

Plot

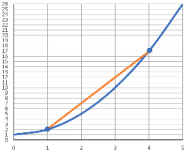


Is Slope same/different on any point on the line?

Functions

- If $f(x) = x^2 + 1$ is function showing distance traveled (y) at each x seconds.
- Secant Slope gives rate of change, ie Speed. and its approximate.
- For different secants, different slopes, so any change in speed is called acceleration.

$$f(x) = x^2 + 1$$



$$m = \frac{17 - 2}{4 - 1}$$

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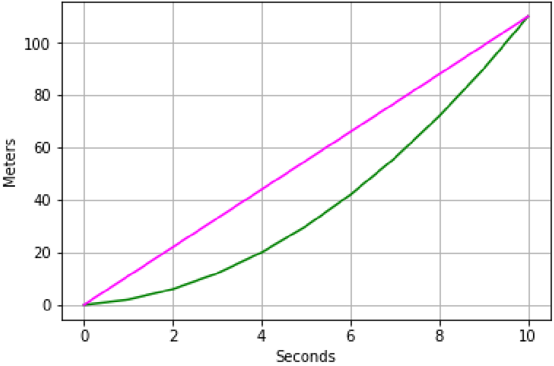
Exercise

$$r(x) = x^2 + x$$

```python
import numpy as np
from matplotlib import pyplot as plt

def r(x):
    return x**2 + x

x = np.array(range(0, 11))
s = np.array([0,10])

plt.xlabel('Seconds')
plt.ylabel('Meters')
plt.grid()

# Plot x against r(x)
plt.plot(x,r(x), color='green')

# Plot the secant line
plt.plot(s,r(s), color='magenta')
plt.show()
```

Plot



Average velocity is 10 m/s

Practical Example

- Say, Pune (P), and Mumbai (M) are 200km apart.
- It takes 4hrs to cover the distance, so the average velocity is $\frac{200}{4} = 50 kmph$.
- Is that the velocity throughout the journey? No. We want more refined answer.
- At Wakad, it could be 20; on highway it could be 100, etc.
- Say, you want to know: within Lonawala, which is about 1km wide-long, whats the velocity?: you can measure from start, end of 1km, and decide.
- Going narrower: What's the velocity while crossing Maganlal shop? which is about 10m long?
- Narrowing down more and more, to find what is called 'instantaneous' velocity.
- Very small, infinitesimally small, a point.
- Thats the "Limiting condition"

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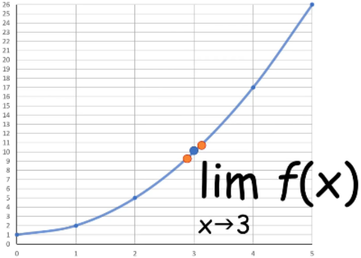
Instantaneous Velocity?

- Secant is an approximation, not an exact speed at a particular moment.
- We need slope of a curve AT A SINGLE POINT (and not approximate average between two secant points)
- If we want slope at a particular x, we can find slope of a secant between x1 and x2 which are very close to x.

Instantaneous Velocity?

- For a point on the curve ,say, (3,10)
- From left and right sides we can approach the point.
- Y value at such nearby points is called as LIMIT
- And its written as:



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The Formal Definition Of A Limit

$$\lim_{x \to c} f(x) = L$$

means for all real ε > 0 there exists a real δ > 0 such that for all x with 0 < |x − c| < δ, we have |f(x) − L| < ε

Math English	Human English
$\lim_{x \to c} f(x) = L$ means	When we "strongly predict" that f(c) = L, we mean
for all real $\varepsilon > 0$	for any error margin we want (+/- .1 meters)
there exists a real $\delta > 0$	there is a zoom level (+/- .1 seconds)
such that for all x with $0 < \|x - c\| < \delta$, we have $\|f(x) - L\| < \varepsilon$	where the prediction stays accurate to within the error margin

(Ref: An Intuitive Introduction To Limits - Better Explained)

A few subtleties

$$\lim_{x \to c} f(x) = L$$

means for all real $\varepsilon > 0$ there exists a real $\delta > 0$ such that for all x with $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$

- The zoom level (delta, δ) is the function input, i.e. the time in the video
- The error margin (epsilon, ϵ) is the most the function output (the ball's position) can differ from our prediction throughout the entire zoom level
- The absolute value condition ($0 < |x - c| < \delta$) means positive and negative offsets must work, and we're skipping the black hole itself (when $|x - c| = 0$).

$$\lim_{x \to c} f(x) = f(c)$$

(Ref: An Intuitive Introduction To Limits - Better Explained)

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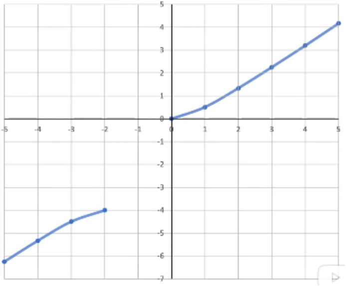
Continuity

- Most functions we looked at so far, are continuous for all real values of x
- Meaning if you trace them, you need not lift your pen.
- Not all functions are defined like that.
- $f(x) = \frac{x^2}{x+1}, x \leq -2 | x \geq 0$
- Obviously, there is a gap in the domain as well as output, so not continuous.



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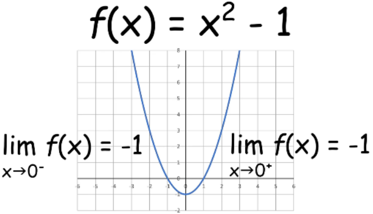
Finding Limits

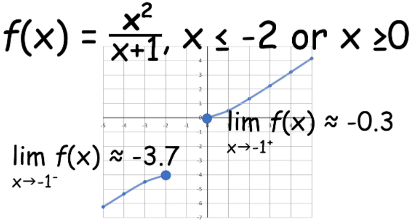
What is meant by finding limits of a function at a particular x?

- Go from left side (ie less of x, increasing) and see to which value $f(x)$ is approaching.
- Go from right side (ie more of x, decreasing) and see to which value $f(x)$ is approaching.
- If both results are same, we got the limit value.
- Even if $f(x)$ is not defined for that x, the limit value exists!!!



Finding Limits

- Lets look at obvious non continuous function
- Need to find limit of function at x = -1
- From negative side, the $f(x)$ is projected to go to 3.7 or so for x = -1
- From positive side, the $f(x)$ is projected to go to -0.3 so for x = -1
- Both do NOT agree. So limit does not exists.



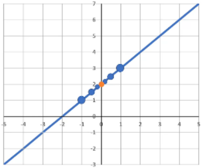
Finding Limits

- Lets look at single point non continuous function
- Need to find limit of function at x = 0
- Calculate values from both sides
- The limit is approaching to 2

$$f(x) = x + 2,\ x \neq 0$$

x	f(x)
-1	1
-0.5	1.5
-0.01	1.99
0.01	2.01
0.5	2.5
1	3

2



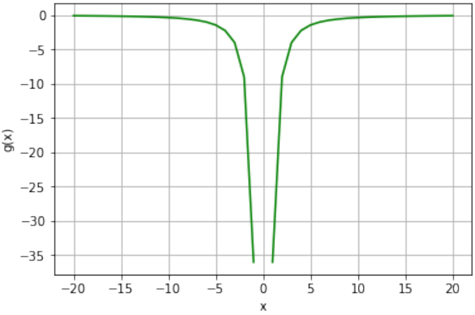
Why not just substitute the value?

Exercise

$$g(x) = -(\frac{12}{2x})^2, x \neq 0$$

```python
from matplotlib import pyplot as plt

def g(x):
    if x != 0:
        return -(12/(2*x))**2

x = range(-20, 21)
y = [g(a) for a in x]

plt.xlabel('x')
plt.ylabel('g(x)')
plt.grid()

# Plot x against g(x)
plt.plot(x,y, color='green')

plt.show()
```

Plot



Exercise

$$d(x) = \frac{4}{x-25}, x \neq 25$$

```
from matplotlib import pyplot as plt

def d(x):
    if x != 25:
        return 4 / (x - 25)

x = list(range(-100, 24))
x.append(24.9) # Add some fractional x
x.append(25)   # values around
x.append(25.1) # 25 for finer-grain results
x = x + list(range(26, 101))
# Get the corresponding y values from the function
y = [d(i) for i in x]

plt.xlabel('x')
plt.ylabel('d(x)')
plt.grid()

plt.plot(x,y, color='purple')
plt.show()
```

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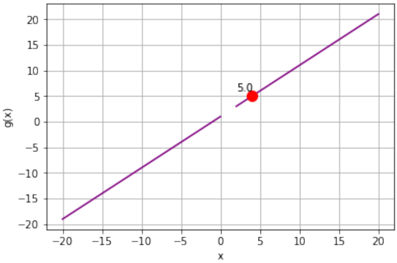
Finding Limits

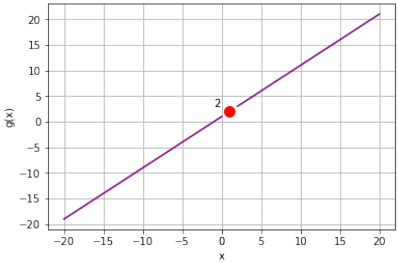
Direct Substitution $g(x) = \frac{x^2-1}{x-1}, x \neq 1$



Find the limit of g(x) as x approaches 4. Just substitute and get the limit as 5.

Finding Limits

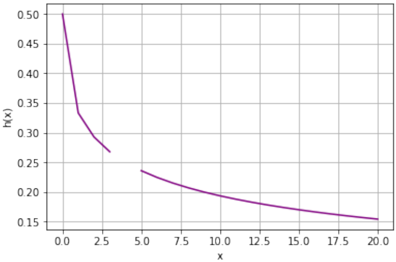
Factorization $g(x) = \frac{x^2-1}{x-1}, x \neq 1$



Find the limit of g(x) as x approaches 1. Substituting gives 0/0, so not good. Factorize and cancel common portion $(x - 1)$. Then substitute. Result is 2.

Finding Limits

Rationalization $h(x) = \frac{\sqrt{x}-2}{x-4}, x \neq 4 and x \geq 0$



Multiply top and bottom by $\sqrt{x}+2$. Then substitute x = 4.. Result is 0.25

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Rules of Limits

- $\lim_{x \to a}(j(x) + l(x)) = \lim_{x \to a} j(x) + \lim_{x \to a} l(x)$
- $\lim_{x \to a}(j(x) - l(x)) = \lim_{x \to a} j(x) - \lim_{x \to a} l(x)$
- $\lim_{x \to a}(j(x) \cdot l(x)) = \lim_{x \to a} j(x) \cdot \lim_{x \to a} l(x)$
- $\lim_{x \to a} \frac{j(x)}{l(x)} = \frac{\lim_{x \to a} j(x)}{\lim_{x \to a} l(x)}$
- $\lim_{x \to a}(j(x))^n = \left(\lim_{x \to a} j(x) \right)^n$

Quiz

True or False:
If a function f is not defined at $x = a$ then the limit $lim_{x \to a} f(x)$ never exists.

Answer

False.
$lim_{x \to a} f(x)$ may exist even if function f is undefined at $x = a$. The concept of limits has to do with the behavior of the function close to $x = a$ and not at $x = a$.

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Quiz

True or False:
$lim_{x \to a} f(x) = +\infty$
and
$lim_{x \to a} g(x) = +\infty$
then
$lim_{x \to a}[f(x) - g(x)]$ is always equal to 0.

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Answer

False.
Infinity is not a number and $\infty - \infty$ is not equal to 0. $+\infty$ is a symbol to represent large but undefined numbers. $-\infty$ is small but undefined number.

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Thanks . . .

- Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
- Call + 9 1 9 8 9 0 2 5 1 4 0 6

(https://medium.com/@yogeshharibhaukulkarni)

(https://www.linkedin.com/in/yogeshkulkarni/)

(https://www.github.com/yogeshhk/)